

PES University

Department of Computer Science & Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #38 — Retail, E-Commerce & Finance
Dropshipping Inventory & Order Orchestrator

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1. Requirements Table

Complete set of Functional Requirements (FR-001–FR-005) and Non-Functional Requirements (NFR-001–NFR-002) for the Dropshipping Inventory & Order Orchestrator.

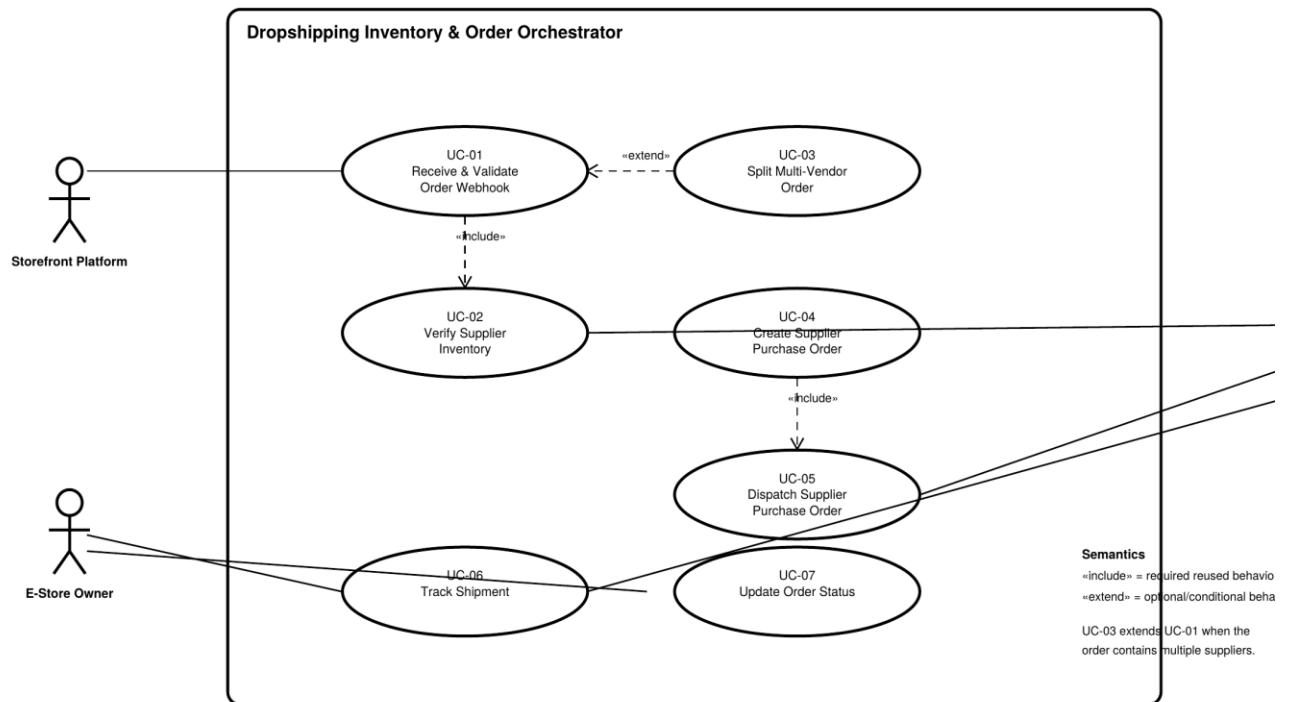
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse and validate inbound order webhooks received from supported storefront platforms such as Shopify or WooCommerce.	High	Pass: A valid webhook containing order ID, customer delivery address, SKU, quantity and order details is parsed and stored successfully. Fail: An invalid or incomplete webhook is accepted as a valid order.	Required to reliably ingest customer orders from the e-commerce storefront.
FR-002	Functional	The system shall verify real-time supplier inventory availability for every SKU contained in an incoming customer order before creating a supplier purchase order.	High	Pass: For each ordered SKU, the system obtains a current supplier availability response before dispatching a supplier PO. Fail: A supplier PO is created for an SKU confirmed to be out of stock.	Prevents supplier orders from being created for unavailable products.
FR-003	Functional	The system shall automatically split an order into separate supplier purchase orders when the order contains products supplied by multiple supplier partners.	High	Pass: An order containing products from two or more suppliers produces one supplier PO for each applicable supplier. Fail: Products belonging to different suppliers are incorrectly combined into a single supplier PO.	A core dropshipping requirement because one customer order may involve multiple suppliers.
FR-004	Functional	The system shall automatically dispatch a supplier purchase order to the appropriate supplier after successful inventory verification.	High	Pass: A valid in-stock SKU results in a supplier PO containing the SKU, quantity and customer delivery address being sent to the correct supplier. Fail: The PO is sent to the wrong supplier or omits required order information.	Automates supplier fulfillment and removes the need for manual order forwarding.
FR-005	Functional	The system shall receive, store and associate supplier tracking numbers with the corresponding customer order and supplier purchase order.	Medium	Pass: When a supplier provides a valid tracking number, it is stored against the correct supplier PO and customer order. Fail: The tracking number is missing, incorrectly associated, or cannot be retrieved for the corresponding order.	Allows the E-Store Owner to monitor fulfillment and shipment status.
NFR-001	Nonfunctional	The system shall acknowledge valid inbound storefront webhooks with HTTP 200 within 200 ms while applying authentication or signature verification to prevent unauthorized webhook requests.	High	Pass: Under simulated peak load, at least 95% of valid webhook requests receive HTTP 200 acknowledgement within 200 ms and requests failing signature verification are rejected. Fail: Valid requests regularly exceed 200 ms or unsigned/invalidly signed requests are accepted.	Fast acknowledgement helps prevent webhook retry storms while verification protects the system from unauthorized requests.
NFR-002	Nonfunctional	The system shall handle temporary supplier API failures without losing customer order information and shall retry failed inventory requests at least three times before marking supplier verification as unsuccessful.	High	Pass: During testing, a temporary supplier API failure causes automatic retries up to three times and the original customer order remains stored. Fail: The order is lost or the system permanently fails after a single temporary supplier API error.	Supplier APIs are external dependencies and may temporarily become unavailable; orders must not be lost.

2. UML Use-Case Diagram

Dropshipping Inventory & Order Orchestrator — Problem Statement #38

UML Use-Case Diagram

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Semantics

«include» = required, reused behaviour.

«extend» = optional / conditional behaviour.

UC-03 extends UC-01 when the order contains multiple suppliers.

3. Use-Case Flow Specification

UC-01: Process Customer Order

Primary Actor	Storefront Platform
Supporting Actors	Supplier Partner; E-Store Owner
Goal	Validate a storefront order, verify supplier stock, and initiate supplier fulfillment.

Preconditions

- The Dropshipping Inventory & Order Orchestrator is operational.
- The storefront platform is configured to send order webhooks.
- Supplier APIs are available and authenticated.
- Supplier product/SKU mappings are configured.

Postconditions

Success: The customer order is stored; supplier inventory is verified; required supplier purchase order(s) are generated and dispatched; and the order status is updated.

Failure: The customer order remains stored, no invalid supplier PO is dispatched, and the failure status is recorded for further handling.

Main Success Scenario

1. The Storefront Platform sends a new customer order webhook to the system.
2. The system acknowledges receipt and validates the webhook.
3. The system extracts the order ID, customer delivery address, SKUs and quantities.
4. The system identifies the appropriate supplier for each SKU.
5. The system requests current inventory availability from the relevant supplier APIs.
6. Each supplier confirms that the requested SKUs are available.
7. The system determines whether the order contains products from multiple suppliers.
8. If required, the system splits the order into supplier-specific purchase orders.
9. The system creates and dispatches the supplier purchase order(s) containing the required SKU, quantity and delivery information.
10. The system records supplier PO information and updates the customer order status to "Fulfillment Initiated".

Alternate Flow A1 — Supplier SKU Out of Stock

- A1.1 At Step 6, a supplier reports that one or more requested SKUs are out of stock.
- A1.2 The system marks the affected SKU as unavailable.
- A1.3 The system prevents a supplier PO from being created for the unavailable SKU.
- A1.4 The system records the inventory failure against the customer order.
- A1.5 The system notifies the E-Store Owner that fulfillment cannot proceed for the affected item.
- A1.6 The order is marked "Inventory Exception" for further action.